

DIVYA PATEL

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EDUCATION

Northeastern University, Khoury College of Computer Sciences Boston, MA

Sep 2025 – Present

Master of Science in Artificial Intelligence

Expected Graduation: May 2027

Related Courses: Foundation of AI, Algorithms, Pattern Recognition & Computer Vision, Applied Data Processing for AI

USCI, Karnavati University, Gandhinagar, India

July 2021 – June 2025

Bachelor of Technology in Computer Science Engineering with specialization in AIML

CGPA: 8.64/10.00

Related Courses: Deep Learning, Natural Language Processing, Applied Machine Learning

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, MATLAB, R

Frameworks & Libraries: TensorFlow, PyTorch, Keras, NumPy, Pandas, sklearn, JAX, Langchain, LangGraph, RAG, MCP, OpenCV

Cloud, DevOps & API: AWS, Azure, GCP, Docker, Kubernetes, Git, CI/CD, Jenkins, CUDA

Computer Vision: ONNX, YOLO, ViT, Depth Anything V2, Haar Cascades, solvePnP, Pillow, NeRF, Gaussian Splatting, ControlNet

Robotics: ROS2, 3D Mapping, Sensor Fusion Algorithms, Multi-Sensor Calibration, Point Cloud Processing, Depth Sensing

Agentic AI: LLM Fine-Tuning, Multi-Agent Systems, Tool Orchestration, RAG, FastMCP, Ollama, Multi-Agent RL

Research Work: Computational Neuroscience, NLP, Computer Vision, Reinforcement Learning, Transfer Learning, Signal Processing

EXPERIENCE

aRoboticsCompany, Inc. (aRC) Ossining, NY

June 2026 – Present

ML/CV Engineering Co-op

- Engineered 3D weld mesh reconstruction using VGGT, Gaussian Splatting, and NeRF, validated against depth point clouds ($\pm 1/32''$).
- Engineered embedded CV pipeline for fatigue crack detection via IR thermography synced over CAN, achieving $\sim 97\%$ precision.
- Designed memory-buffer algorithm (0.08s offset) for real-time crack detection via acoustic-induced heat anomaly isolation.
- Built RGB-IR pipeline using stereo calibration and cross-spectral fusion for 16-bit thermal inference to detect leaks behind walls.
- Optimized inference pipelines using ONNX Runtime to enable low-latency, real-time performance on edge hardware.

NEUMove Lab, Northeastern University, Boston, MA

Oct 2025 – May 2026

Research Assistant

- Analyzing gait samples using biomechanical signals and ML models to predict center-of-mass dynamics.
- Architected Bidirectional LSTM models for time series analysis, improving prediction accuracy by 15%.
- Extracted center-of-mass trajectories to evaluate gait stability across 10+ biomechanical conditions.

Indian Institute of Technology Gandhinagar, Maker Bhavan Lab  Gandhinagar, India

May 2024 – July 2024

Innovator, Machine Learning and Product Development Intern

- Designed SaMD neurostimulation device for personalized treatment of Willis-Ekbom Disease.
- Trained LSTM on 10,000+ synthetic EMG signals with signal processing pipelines, achieving 75% prediction accuracy.
- Prototyped hardware using PCB design, 3D printing, ESP32, EMG processing, and CAD tools (KiCad, SolidWorks).

BrainyBeam Info-Tech  Ahmedabad, India

July 2024 – Sep 2024

Data Science and Machine Learning Intern

- Built fuel consumption prediction system using Linear Regression, delivering R^2 score of 0.45.
- Preprocessed 5,000+ records using NumPy, Pandas, and Scikit-learn, reducing data cleaning time by 30%.
- Visualized key variable relationships using Matplotlib and Seaborn, evaluating performance with MAE, MSE, and RMSE metrics.

PROJECTS

Darwin: Privacy-Focused AI Research Agent 

Jan 2026 – Apr 2026

- Built local agentic research assistant using Ollama and FastMCP, connecting arXiv, PDF parser, and Obsidian tool servers.
- Designed multi-turn LLM reasoning loop with MCP tool routing, schema normalization, and human-in-the-loop approval gates.
- Achieved 0.94 category precision on arXiv search, 98% PDF text recovery, and sub-100ms extraction latency across papers.

Calibration & Augmented Reality 

Feb 2026 – Mar 2026

- Calibrated camera intrinsics using checkerboard detection across 20+ frames, achieving sub-pixel reprojection error.
- Estimated real-time pose via solvePnP and overlaid 3D virtual objects onto live video with accurate perspective.
- Extended pipeline with Harris corner detection and occlusion polygon rendering for depth-aware AR at real-time rates.

Multi-Agent Aerial Dog Fighting

Sep 2025 – Dec 2025

- Developed scalable MARL framework for 6 UAV agents using Ray RLlib and PPO algorithm.
- Generated interactive 3D simulations analyzing 1,000+ training episodes for trajectory optimization.
- Created modular architecture separating environment, reward logic, and model configuration.

Fine-Tuning LLMs for Domain-Specific Applications 

Jan 2024 – Apr 2024

- Fine-tuned 5 LLMs (Mistral, LLaMA2, Idefics, BERT, Gemma) for legal, medical, and customer service domains.
- Applied LoRA, AdamW, and FP16 optimization techniques, reducing computational overhead by 40%.
- Deployed on AWS/Azure achieving sub-100ms latency with 20–30% improvement in domain-specific accuracy.

PUBLICATIONS AND PATENT

Advanced Public Bus Transport Management System: An Innovative Smart Bus Concept

Feb 2024

Presented at [42nd IEEE ICCE](#) — IEEE Xplore

Provisional Patent: Device and Method for Treatment of RLS

July 2024

India App. No. 202411083538 — USA App. No. 63/818,733